Robot Path

Time limit: 1.00 s **Memory limit:** 512 MB

You are given a description of a robot's path. The robot begins at point $(0, 0)$ and performs n commands. Each command moves the robot some distance up, down, left or right.

The robot will stop when it has performed all commands, or immediately when it returns to a point that it has already visited. Your task is to calculate the total distance the robot moves.

Input

The first line has an integer n: the number of commands.

After that, there are n lines describing the commands. Each line has a character d and an integer x: the robot moves the distance x to the direction d. Each direction is U (up), D (down), L (left), or R (right).

Output

Print the total distance the robot moves.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq x \leq 10^6$

Example

Input	Output
5 U 2 R 3 D 1 L 5 U 2	9